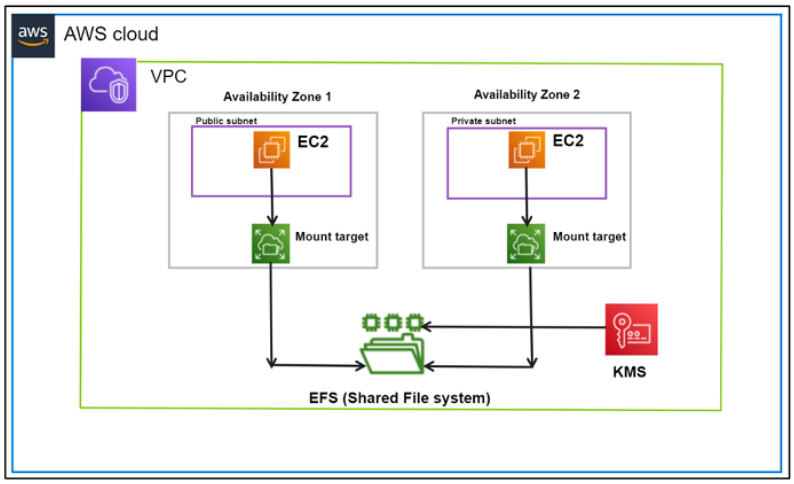
AWS Assignment Case Study 2

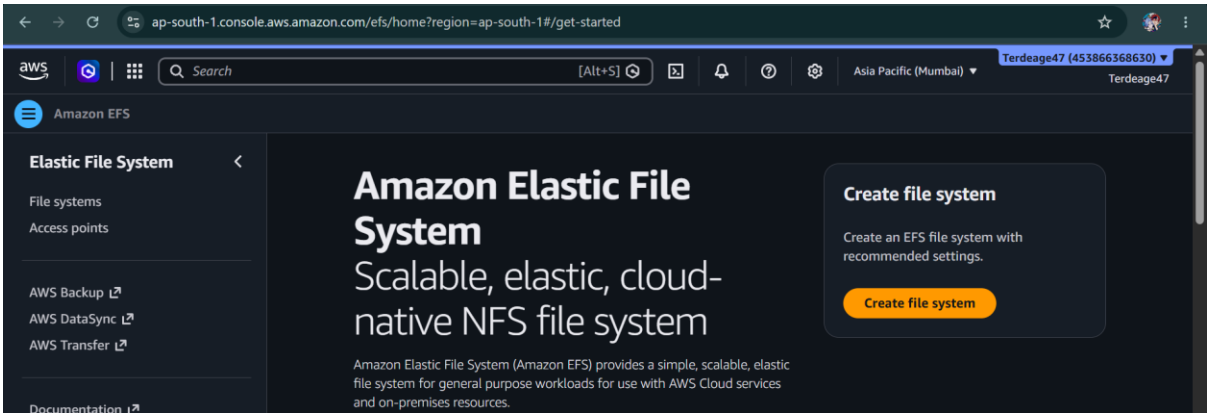
Shared File System using AWS EFS

Problem Statement ABC is India based entertainment production company with a focus on Northeast and East Indian cinema. They wanted a highly available and reliable storage solution for their on prem data. The client wanted the data to be reliable, highly available, secure, and persistent and also wanted to have data on the storage to be accessible from all the servers. The issues with the existing infrastructure are mentioned below: 1) They were using NAS storage which had limitations in scalability. 2) The client wanted a complete application to run in Aws cloud with storage in the cloud to keep his files in sync with servers so that he has centralized storage of data 3) There were security issues in the existing infrastructure and no encryption at rest or transit. 4) Manual intervention was needed to change the storage type and transfer files to infrequent storage. 5) The client is unable to scale up the infrastructure due to high capital costs for new hardware. 6) Need for low-cost storage options for both frequent and infrequent data. 7) Highly available, secure, and persistent shared File system in AWS cloud with EFS.

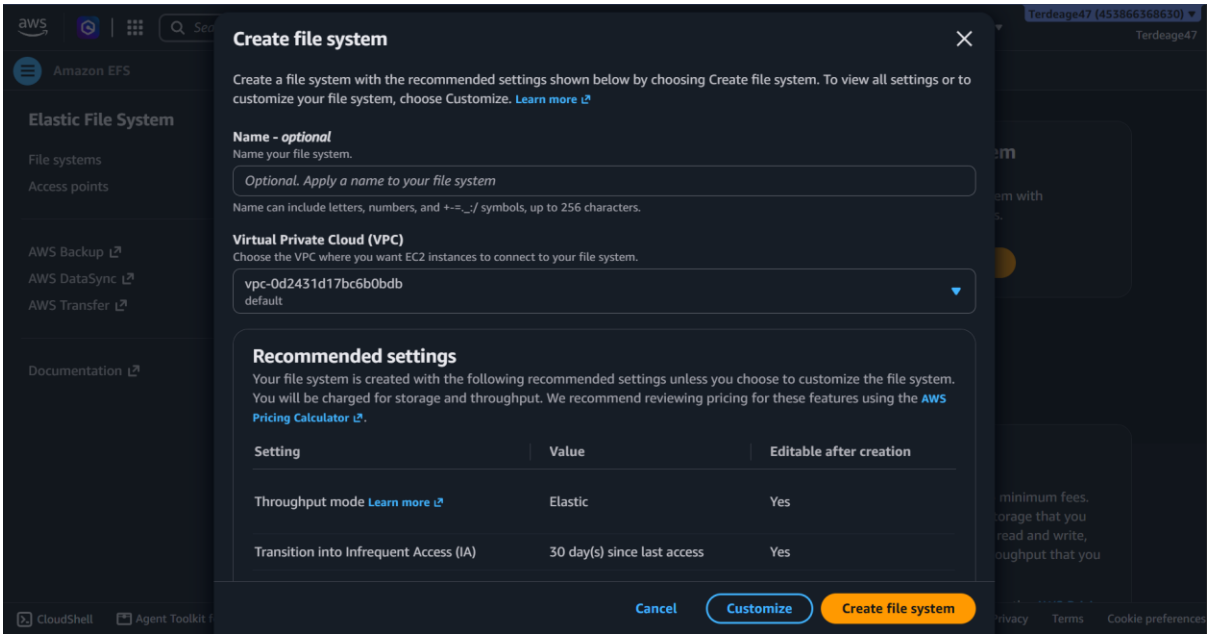
Architectural Diagram



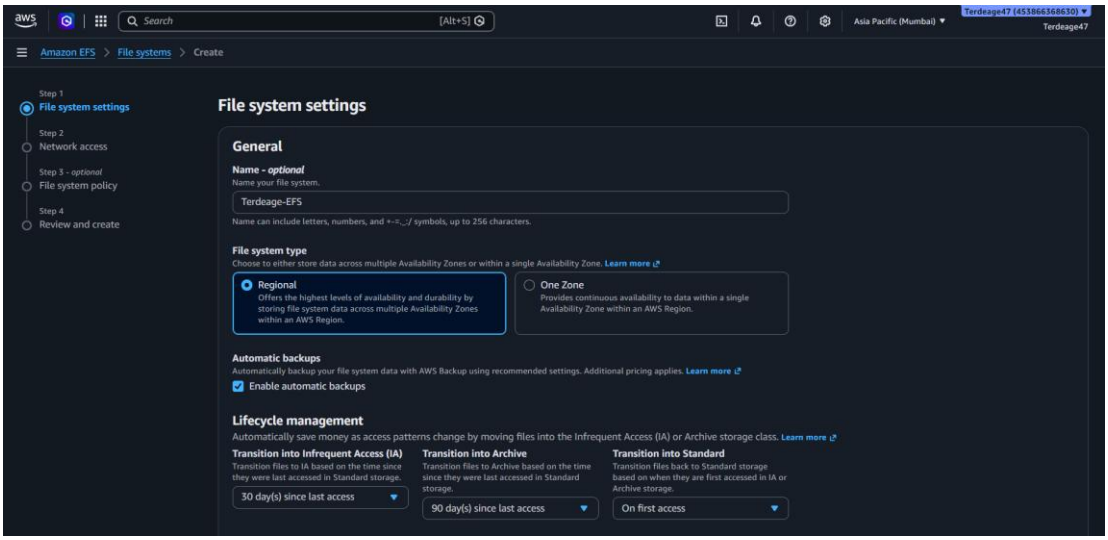
1. Search for EFS in Amazon console



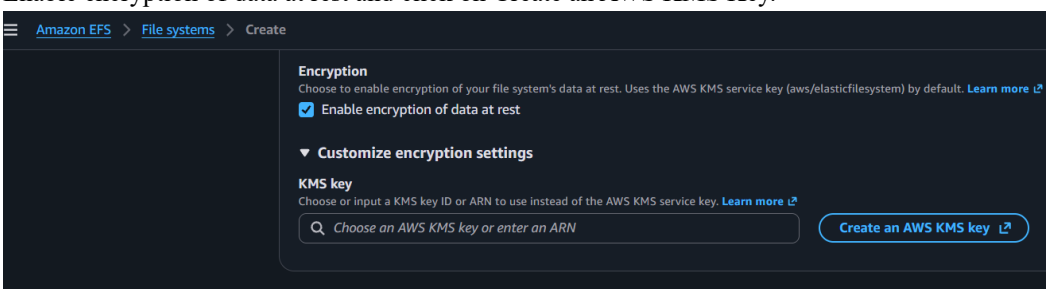
2. Click on Create file system, a popup will open, click on customize.



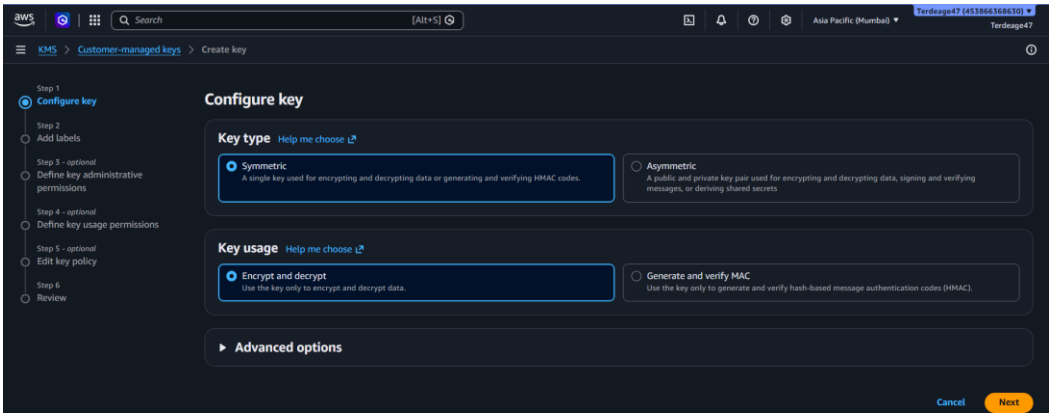
3. Enter the respective data as below:
   a) Name = Name of your choice
   b) Storage class = standard/ regional
   c) Automatic backup = enabled
   d) Transition to IA = Choose 30 days from the list
   e) Transition out of IA = Choose On first access



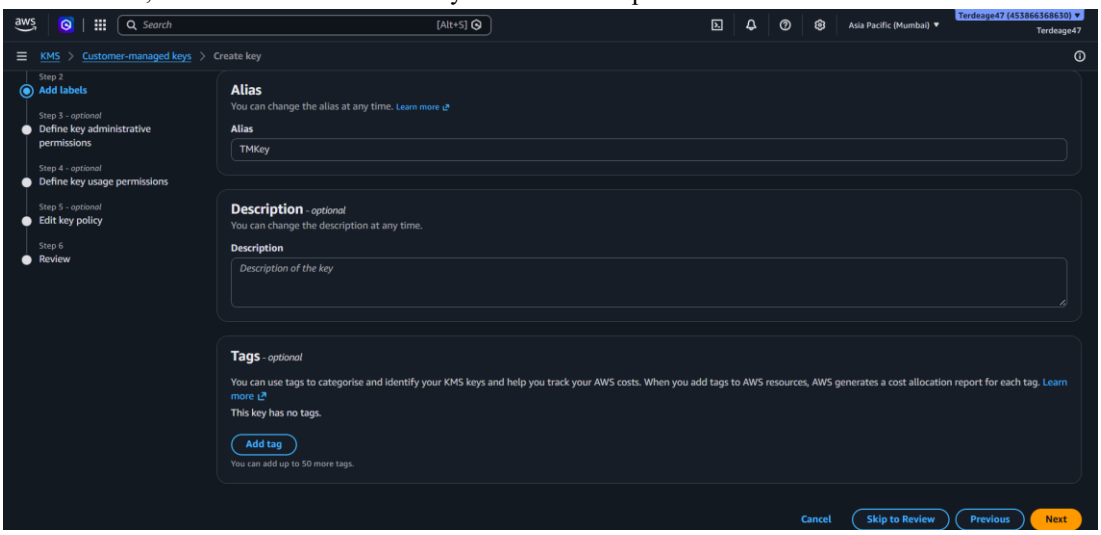
4. Enable encryption of data at rest and click on Create an AWS KMS Key.



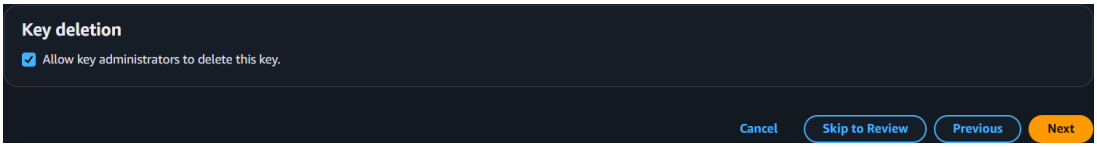
5. After the KMS dashboard is open. Enter the respective data as below:
   a) Key type = Symmetric
   b) Key usage = Encrypt and decrypt
   c) Keep rest as default and click Next.



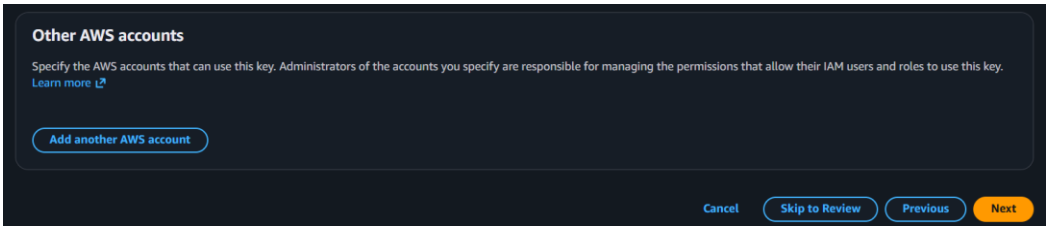
In Add labels, Enter Alias with a name of your choice. Keep rest as default and Next.



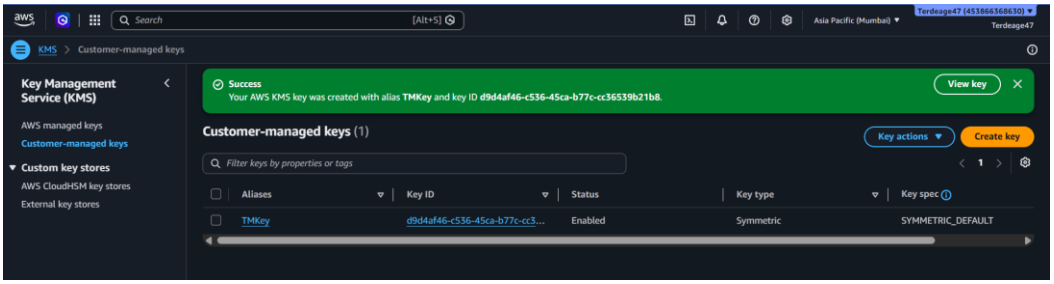
Leave key administrator and key deletion options as default and click Next.



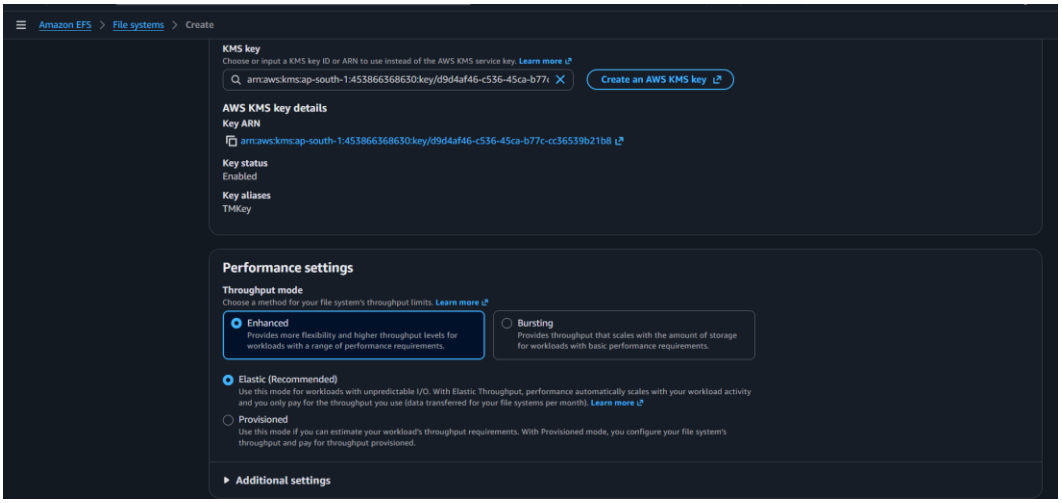
Keep Key usage permissions as default and click Next.



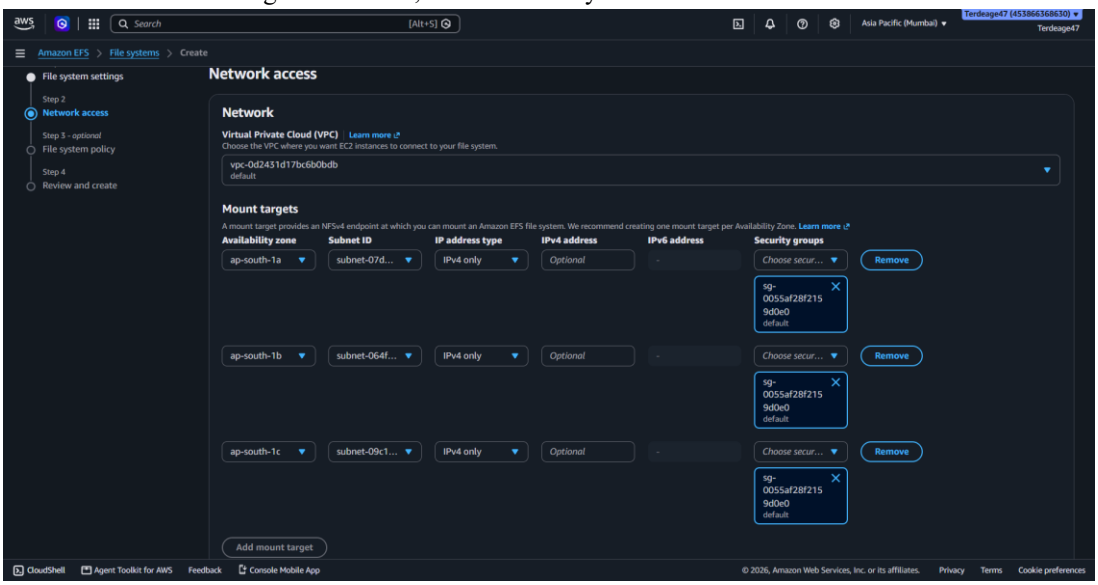
Review and Click on Finish.



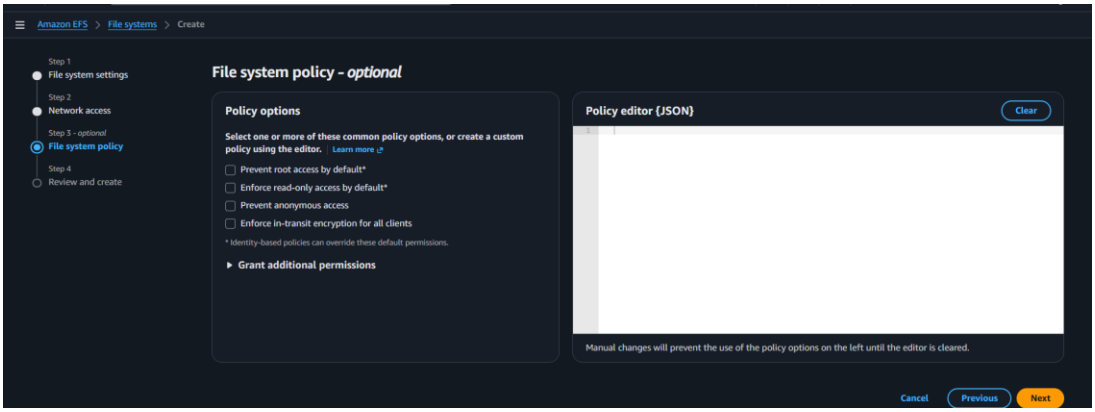
6. Go back to EFS create wizard screen and Choose the KMS key created in previous step from drop down list. Leave rest as default and then Next.



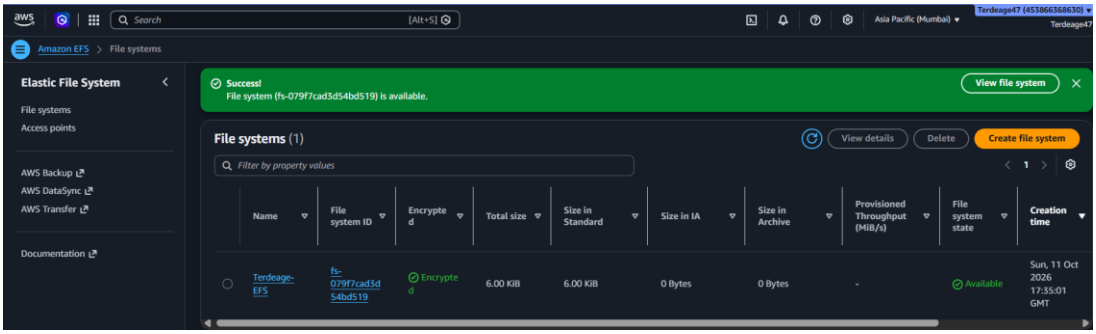
In Network access setting. Under VPC, Select VPC of your choice. Leave rest as default and click Next.



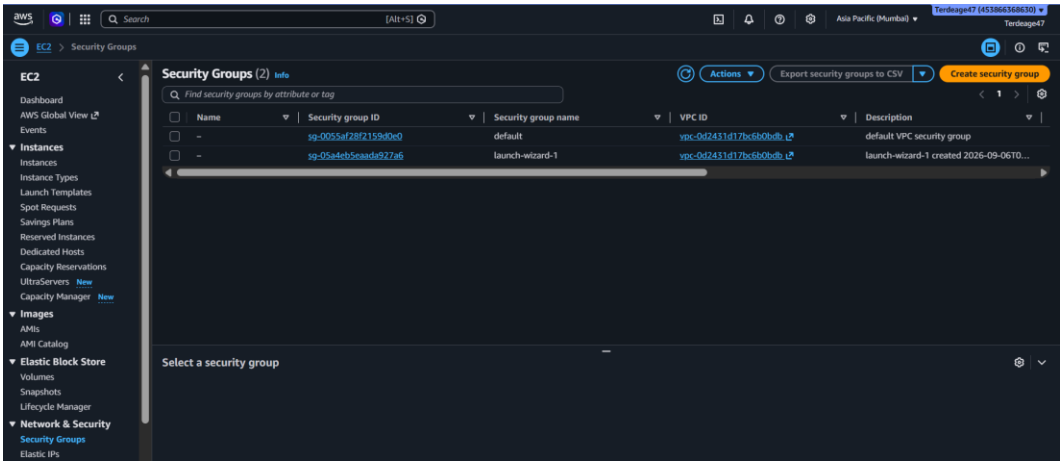
In File system Policy leave as default and Click on Next.



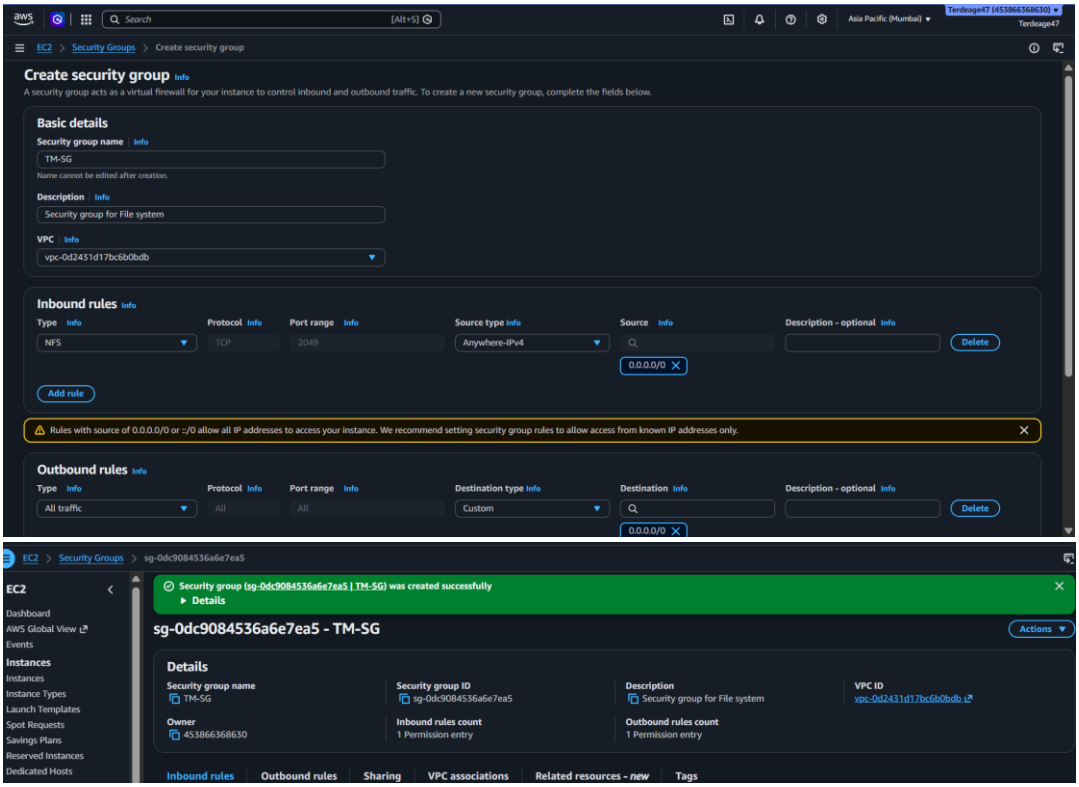
Review and Click on Create.



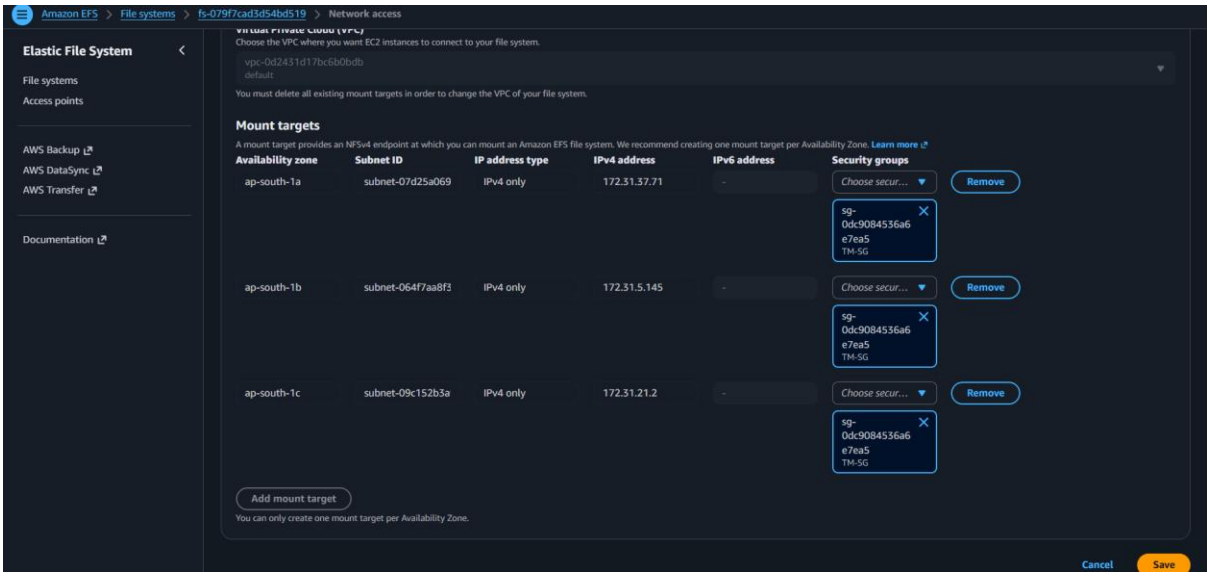
7. Search for EC2 in Amazon console and go to Security Groups under Network and Security settings. Click on Create Security Group



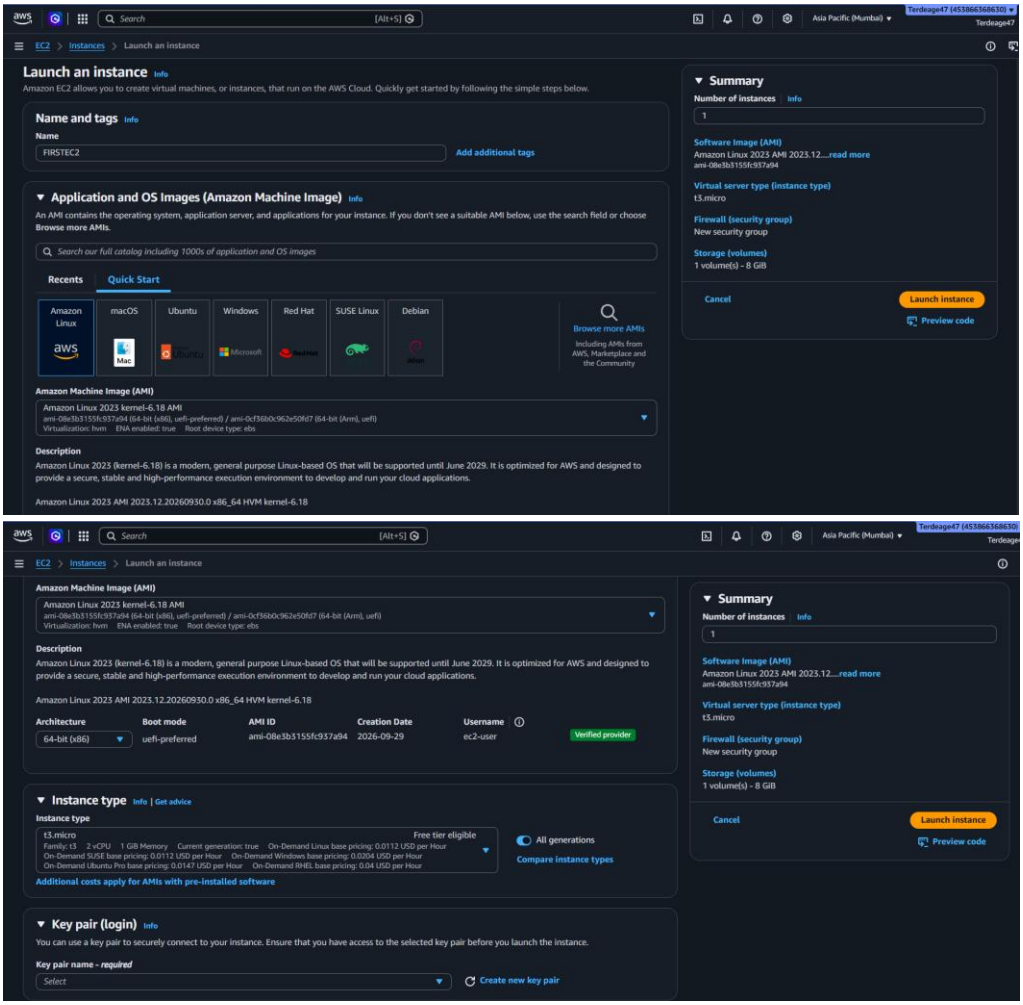
8. Create Security group wizard will open. Then enter the following details:
   a) Security group name = Name of our choice
   b) Description = Security group for file system
   c) VPC = Give ID of the VPC in which file system is created.
   In Inbound rule click on Add rule. Select Type = NFS and Source = IPV4 anywhere. Leave rest as default and Click Create security group.



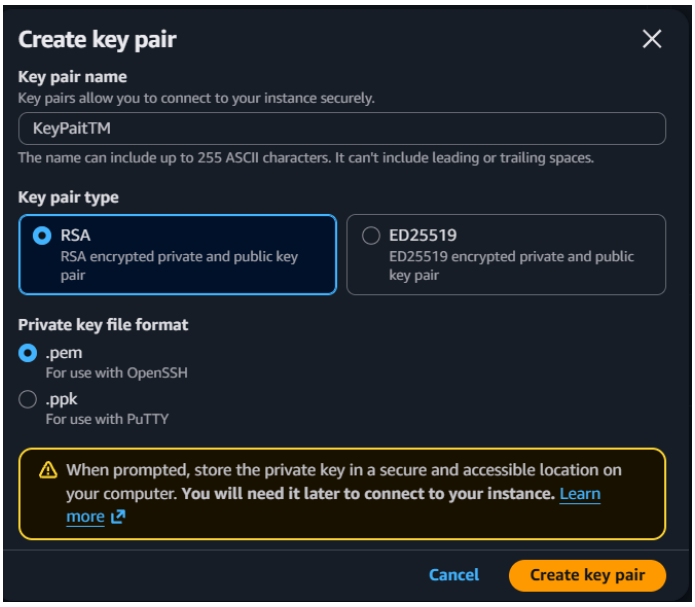
9. Go back to EFS. Select your file system -> Click on View details. Click on Network Tab in details page. Under Network Tab -> Select Manage option. In each Mount target -> remove default security group -> add security group created in previous steps with NFS protocol. Click on Save.



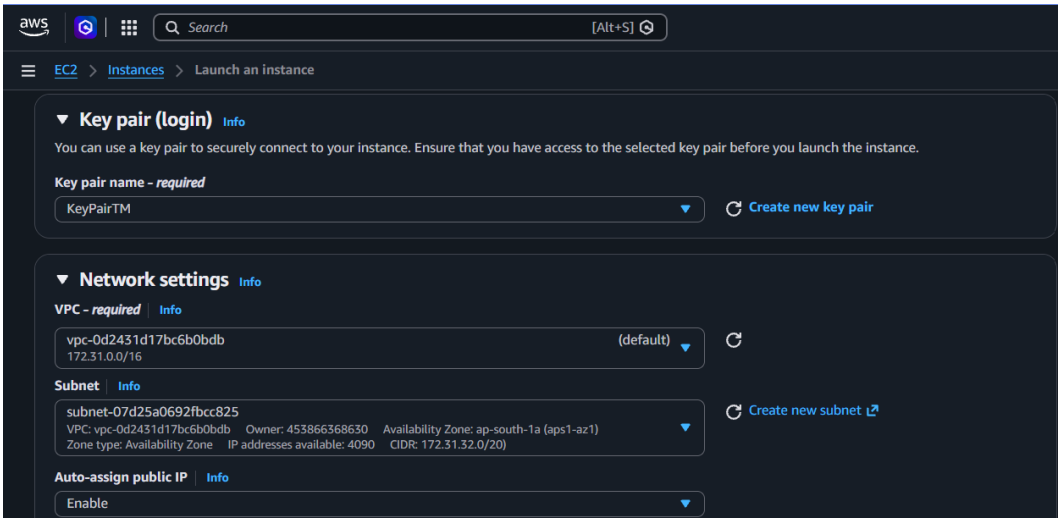
10. Go back to EC2, Click on Instances. Then Launch Instances.
    Launch wizard will open, then enter the following:
    a) Name = FIRSTEC2
    b) Number of instances = 1
    In application and OS image -> Amazon Linux
    In Instance type -> Select t3.micro



In Key pair -> click on Create new keypair. In Create key pair tab, Enter Name of your choice, leave rest as default and click Create key pair.



In network settings -> Click on Edit.
Select a) VPC = VPC in which file system is created
b) Subnet = Choose subnet of your choice
c) Auto assign Public IP = Enable



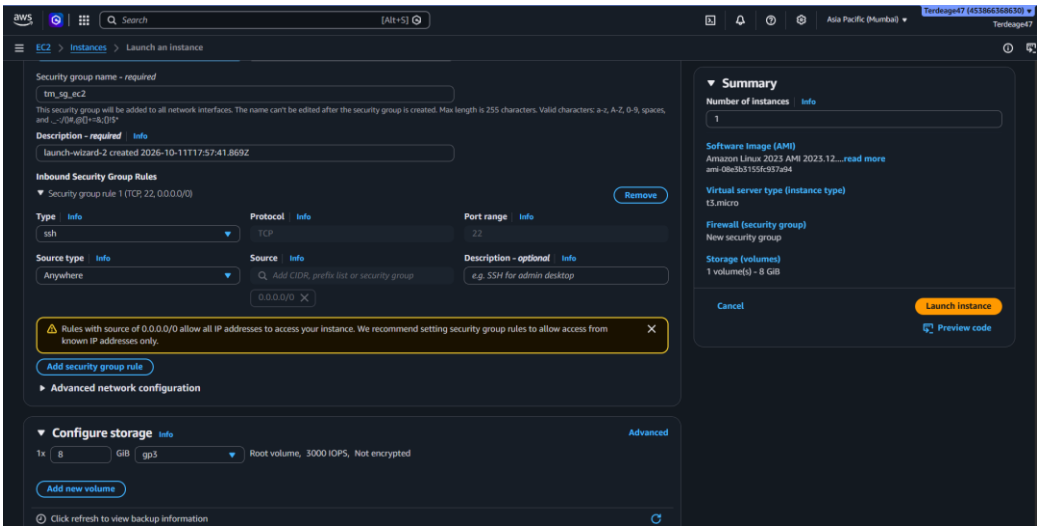
Select Create Security group and Enter

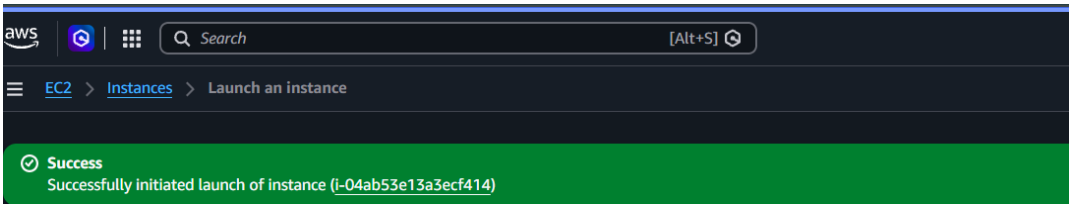
a) Security group name = Name of your choice

b) Type = SSH

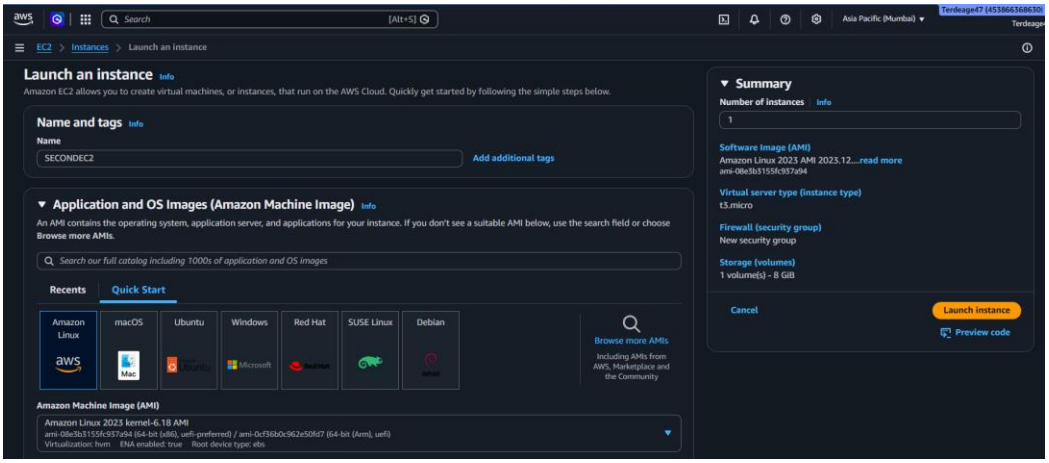
c) Source = IPV4 anywhere

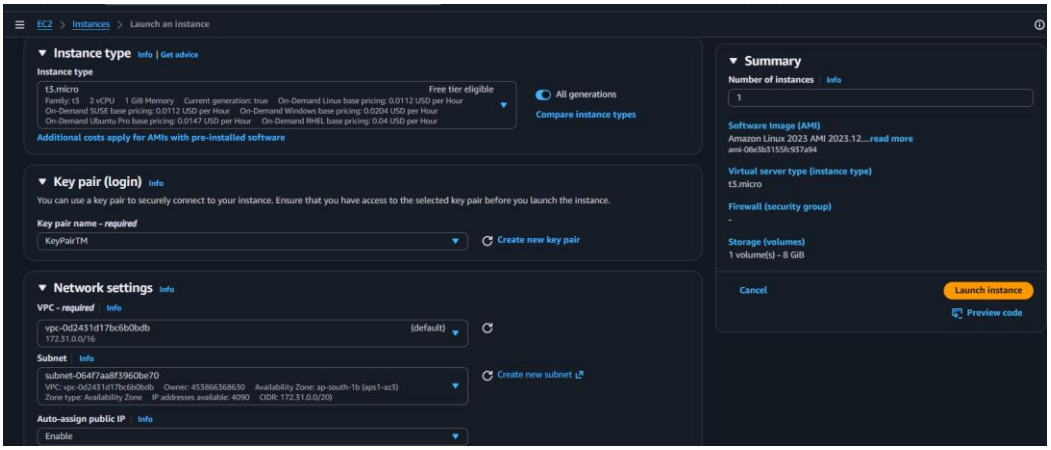
Leave rest as default -> Click Launch Instance.

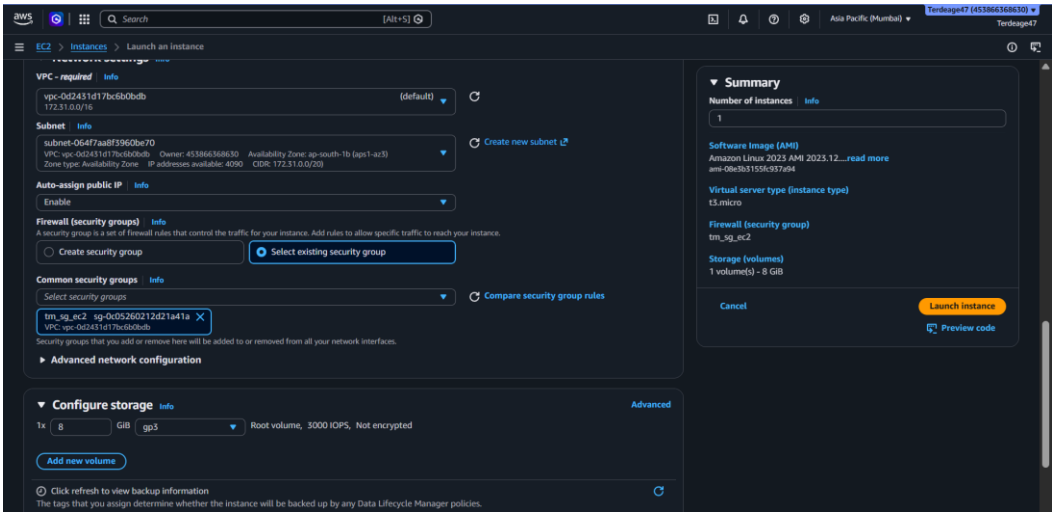




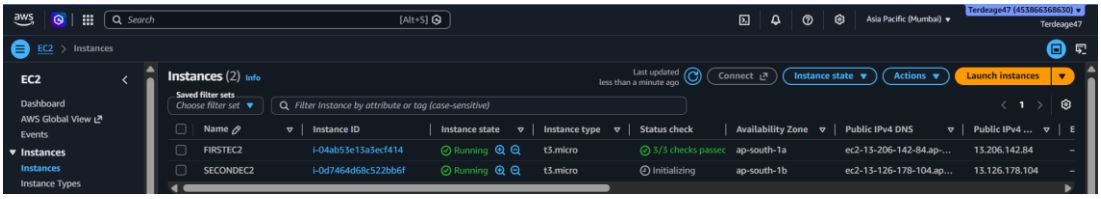
11. Launch one more EC2 repeat step number 10, In this EC2 give
    a) Name = SECONDEC2
    b) Subnet = choose subnet in availability zone different from firstEC2



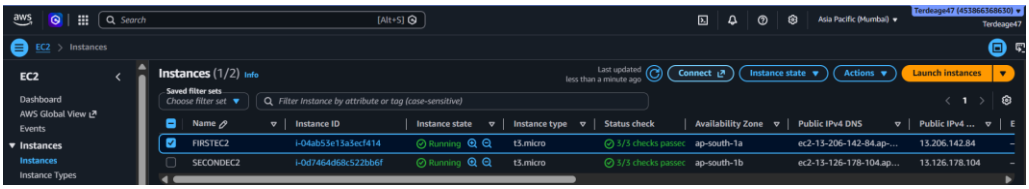




Observe both EC2 instances are launched in different availability zones.



12. Select FIRSTEC2 instance -> click CONNECT.



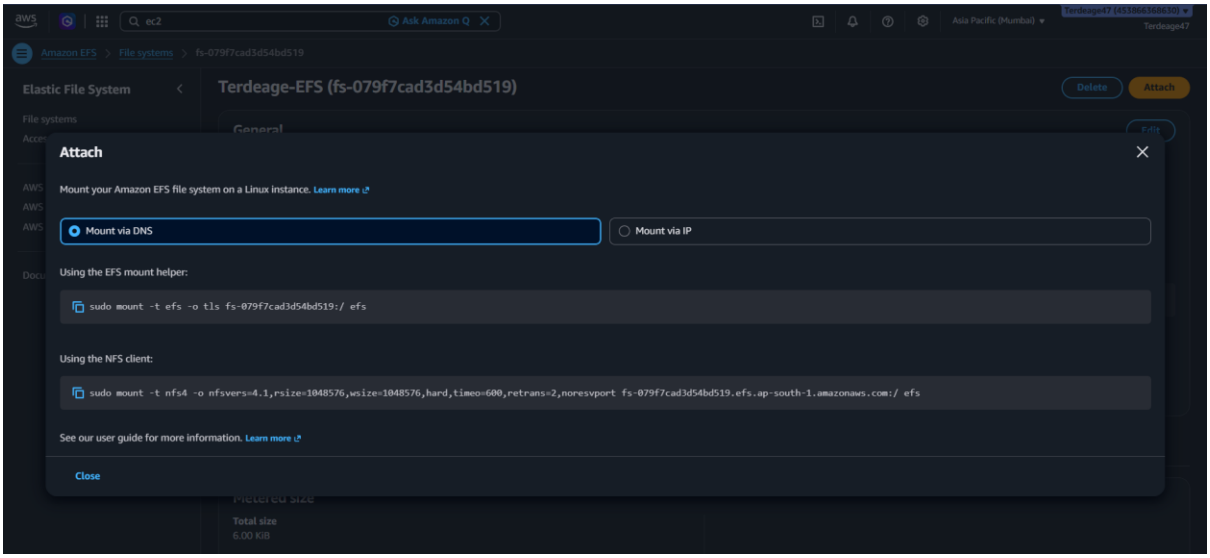
Select EC2 Instance connect -> click Connect.

Now run the following commands:

sudo yum install amazon-efs-utils
mkdir efs



13. From services menu -> open EFS in new tab. Select your EFS filesystem -> Click on View details Click on Attach.

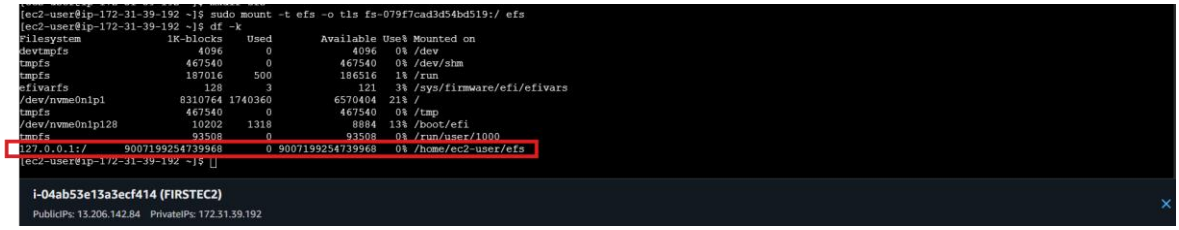


Select Mount via DNS -> copy first command (This command has option parameter as tls which will enable encryption in transit).

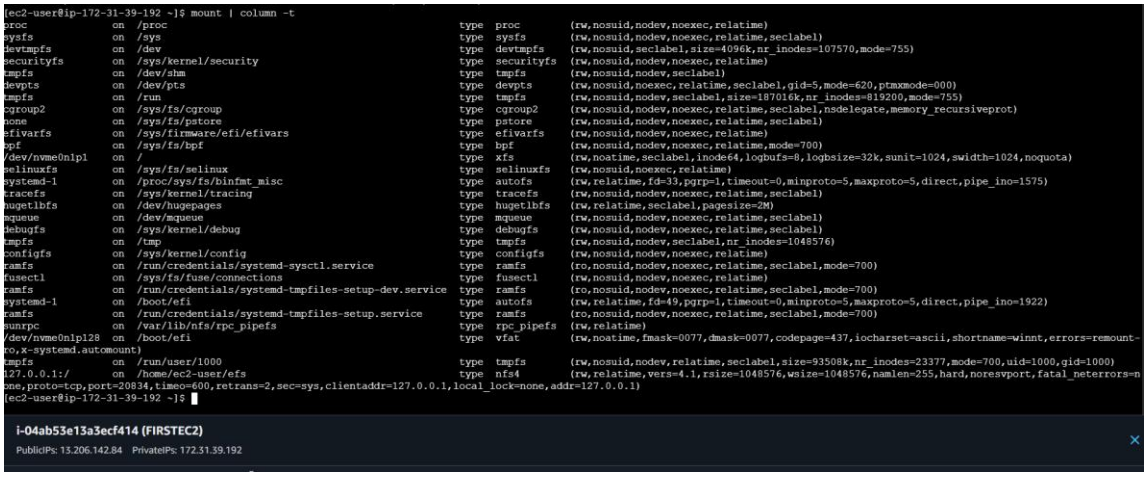
Paste the command in EC2 CLI and press enter.



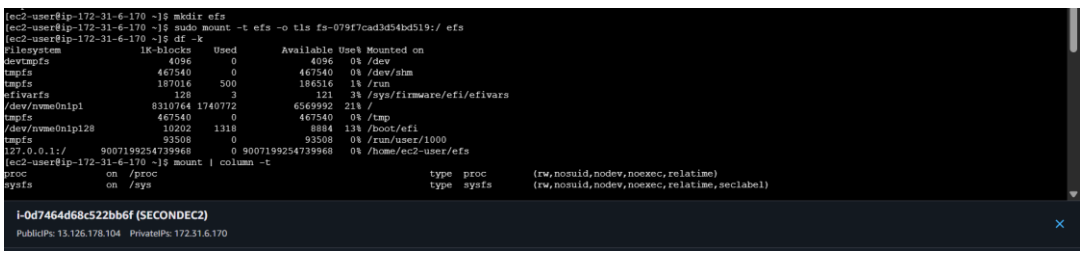
Enter command $ df-k and notice filesystem is mounted on /home/EC2-user/efs directory.

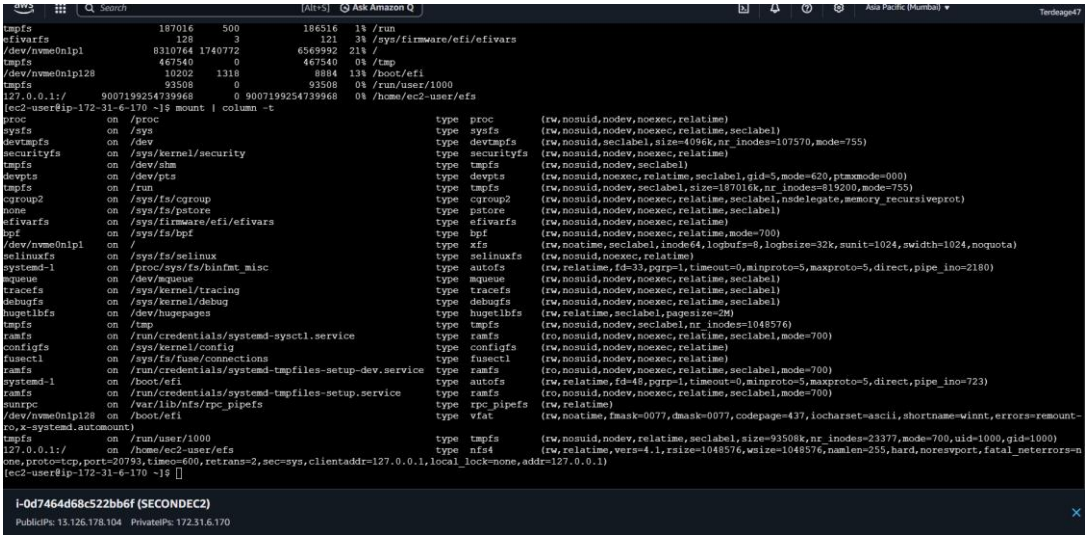


Check if encryption at transit is enabled or not by giving command
$ mount | column -t

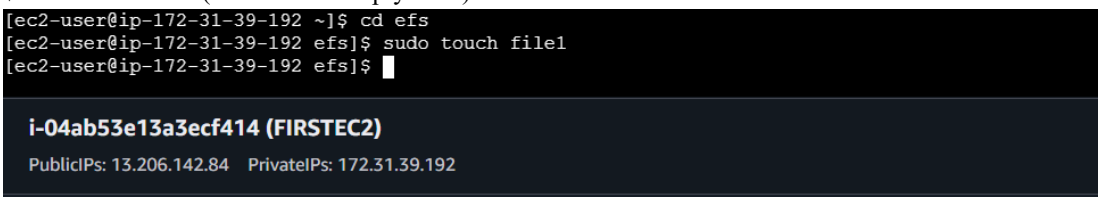


14. Now connect to SECONDEC2 and mount filesystem on it. REPEAT step number 13. And notice that file system is mounted on SECONDEC2.





15. Now in FIRSTEC2 CLI -> Enter command
    $ cd efs (to go inside efs directory)
    $ sudo touch file1 (to create an empty file1)



16. Now go to SECONDEC2
    $ cd efs
    $ ls (to list all contents of directory)



    We can access file1 created by FIRSTEC2 through SECONDEC2. Hence, shared file system is working with both EC2.

17. In EC2 dashboard select both instances -> Instance state -> Terminate Instance. In EFS dashboard -> Select your file system -> click Delete